

Technical, economic and environmental assessment of household biogas digesters for rural communities



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ABSTRACT

This study was carried out in response to the growing interest on household biogas digesters in Latin America, particularly in rural Andean communities. The aim was to compare the fixed dome and plastic tubular digester in terms of biogas production, cost and environmental impact, using the life cycle assessment methodology. Design and operational parameters, construction materials and implementation costs were based on our previous research and literature results for plastic tubular and fixed dome digesters, respectively. According to this analysis, the main advantage of the plastic tubular digester was its ease of implementation and handling, and lower investment cost compared to the fixed dome digester, which appeared to be more environmentally friendly.

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1. Introduction

Household biogas plants have been used for several decades in rural areas of Asian countries like China and India [1–4]. The most widely used household systems are the Chinese fixed dome digester, the Indian floating drum digester and the Taiwanese plastic tubular digester [5–7]. The most appropriate design depends on weather conditions and socio-economic context [6,8].

In Latin America, household digesters have been spreading since the 1980s in rural areas of tropical countries like Colombia and Costa Rica [9–11] and hilly regions of Peru and Bolivia [12,13]. The first experiences on digesters implementation at the Peruvian Andes date back to the 1980s, when the Universidad Nacional de Cajamarca (UNC) together with the NGO INTITEC carried out a project on the implementation of Chinese fixed dome digesters

[14–16]. In 1988 there were almost one hundred fixed dome digesters of 10–12 m³ implemented in rural zones of the Department of Cajamarca [14,15]. Basic objectives of the biogas program were [15,17]: (i) to generate energy for cooking and lighting, satisfying household needs; (ii) to avoid deforestation and its effect on soil erosion; (iii) to reduce environmental pollution; and (iv) to improve household sanitation. When the program started there was a lot of interest in this novel technology, but after a few years people abandoned the digesters and the implementation of new household biogas plants turned off. A viability analysis pointed out that the program failure was mainly due to the high capital cost of fixed dome digesters; but also to social aspects, such as the lack of an appropriate management, family training and project monitoring [15].

During the last decade, there has been a growing interest in implementing plastic tubular digesters in rural Andean communities; because of its low-cost, and ease of implementation and handling [12]. In 2007 Peruvian and international NGOs started a project dealing with the implementation of plastic tubular digesters adapted to rural Andean communities in the Department of Cajamarca [18,19]. The main goal was to improve the living quality of rural families, by providing a clean fuel to replace traditional biomass. The project also aimed to: (i) preserve the environment by reducing greenhouse gas (GHG) emissions and deforestation; (ii) decrease families expenses for fuel and

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fertilizer; and (iii) reduce the workload and time spent by women and children for firewood collection [19]. A recent survey evaluated plastic tubular digesters technical, environmental and socio-economic impacts on rural communities of the Department of Cajamarca. The results showed how digesters improved household living conditions and economy, while reducing GHG emissions compared to traditional biomass [19]. However, the benefits were restricted by a poor performance of anaerobic digestion at high altitude. Indeed, under the harsh weather conditions of the Andean Plateau, digesters require proper insulation and longer hydraulic retention time (HRT) compared to analogous digesters implemented in the Tropics, where they were originally developed; which may increase the capital cost. From an environmental point of view, the eventual use of large amounts of plastic is a matter of concern. Optimizing the performance, economics and environmental sustainability of household digesters in rural Andes is currently considered a strong challenge [19].

The aim of this research was to compare the performance, cost and potential environmental impact of fixed dome and plastic tubular digesters implemented at high altitude.

2. Material and methods

2.1. Household digesters performance

The study considered two types of household digesters adapted to Andean conditions: 1) fixed dome – Chinese model and 2) plastic tubular – Taiwanese model (Fig. 1). Digesters performance comparison was based on a thorough literature review of case studies dealing with household digesters implementation at high altitude (Table 1).

Chinese fixed dome digesters (Fig. 1a) consist of a cylindrical chamber, a feedstock inlet and a digestate outlet, which also serves as a compensation tank [21]. Biogas is stored in the upper part of the chamber. When biogas production starts, the slurry is displaced into the compensation tank. The volume of the compensation tank

is equal to the volume of biogas storage. Gas pressure increases with the volume of biogas stored and with the difference of slurry levels between the inside of the digester and the compensation tank. It is completely underground, and made with bricks and concrete [5,22].

In plastic tubular digesters (Fig. 1b), wastewater flows through a tubular polyethylene or PVC bag (the reactor) from the inlet to the outlet, while biogas is collected by means of a gas pipe connected to a reservoir. Because the system is unheated, in cold hilly regions the plastic bag is buried in a trench and covered by a greenhouse, aimed at increasing inner temperature and reducing temperature fluctuations. Design criteria for the digester, trench and greenhouse depend on each location; at high altitude long HRT of 60–90 days are generally used [12]. Biogas is stored in a reservoir, connected to the kitchen or cooking area.

2.2. Economic analysis

The economic analysis was based on the digesters capital cost, which accounts for the major cost of household biogas plants. Design parameters of digesters implemented in Peru were adopted for the Chinese fixed dome [15,23] and plastic tubular [12] digesters (Table 2). Operational parameters (Table 2) were also representative of household digesters implemented at high altitude [8,24]. Construction materials for each digester are summarised in Table 3. The considered lifespan was 20 years for all materials, except for plastic which was reduced to 5 years according to manufacturers' specifications and Nzila et al. [21].

2.3. Life cycle assessment

Life cycle assessment (LCA) is a methodological tool for the evaluation of environmental aspects and potential impacts through the whole life of a product, process or service; from raw materials extraction, to utilization and final disposal. In brief, LCA comprises mass and energy balances applied to the system, plus an assessment of the environmental impacts associated to the inputs and outputs. According to the ISO 14040 [25], there are four main stages in LCA: goal and scope definition, inventory analysis, impact assessment and interpretation of the results.

2.3.1. Goal and scope definition

The goal of this LCA model was to compare the environmental impact of both household digesters, 1) fixed dome – Chinese model and 2) plastic tubular – Taiwanese model, implemented at high altitude.

The main function of the system was to produce biogas for cooking; therefore the functional unit was defined as the production of 1 m³ of biogas (under standard conditions).

The system analysis included raw materials for household biogas plants construction and maintenance. Bearing in mind that both of them operated under the same conditions, the same biogas production rate was assumed (0.12 m³biogas m⁻³digester d⁻¹) [24]. Biogas combustion for cooking and digestate reuse in agriculture were excluded from the system boundaries, since they were assumed to cause the same impacts in both scenarios.

2.3.2. Inventory analysis

Inventory data on digesters design and operation were the same as for the economic analysis, as the same construction materials and lifespan were considered (Tables 2 and 3). Household digesters were then described and quantified with reference to the functional unit (Table 4). Data concerning the embodied environmental aspects of materials were taken from Ecoinvent v2.2 database [26].

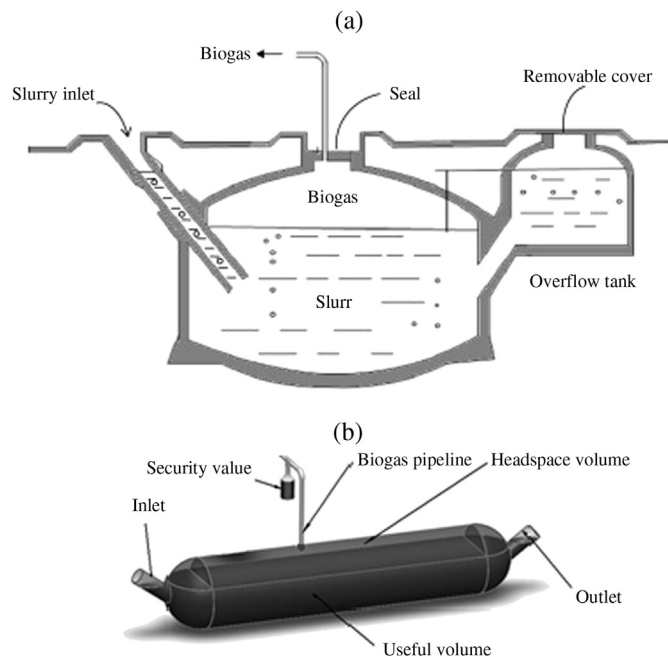


Fig. 1. Schematic diagram of the studied household digesters: (a) fixed dome – Chinese model [20] and (b) plastic tubular – Taiwanese model [12].

Table 1
Performance of small-scale digesters at high altitude.

Reference	Location and m.a.s.l (m)	Digester design	Temperature (°C)	V (m ³)	HRT (d)	Feedstock	OLR (kg VS m ⁻³ digester d ⁻¹)	Biogas production rate (m ³ biogas m ⁻³ digester d ⁻¹)	Methane (% CH ₄)
Ferrer et al. [12]	Peru 2800–3400	Tubular polyethylene or PVC	<25	2.4–7.5	60–90	Cow manure	0.22–1.29	0.07–0.47	65
Garfi et al. [31]	Peru 2800	Tubular PVC	<25	7.5	75	Guinea pig manure	0.59	0.04	60
Garfi et al. [24]	Peru 2800	Tubular PVC	<25	7.5	60–90	Cow and guinea pig manure	0.34–1	0.12 (cow) 0.03 (guinea pig) 0.08 (codigestion of cow and guinea pig)	55–60
Kanwar and Guleri [8]	India 1300	Tubular neoprene Fixed dome (bricks)	<25	1.2	55	Cattle dung	<1	0.50 (tubular neoprene) 0.68 (fixed dome)	55
Kalia and Kanwar [29]	India 1300	Fixed dome	<25	2.4	55	Cattle dung	<1	0.4–0.7	55–60
Kanwar et al. [28]	India 1300	Fixed dome	<25	2	55	Cattle dung	<1	0.35–0.45	55–60
Tasdemiroglu [30]	Turkey 1000	Fixed dome	<25	9.5	45	Cattle and sheep dung	0.38	0.34–0.5	60

Note: HRT: hydraulic residence time; OLR: organic loading rate; V: useful volume.

2.3.3. Impact assessment

Potential environmental impacts were evaluated using the software SimaPro[®] 7.3 (PRé Consultants) and Ecoinvent v2.2 database [26]. This analytical tool is in accordance with ISO 14040 standards [27]. The following impact categories were evaluated: Abiotic Depletion (AD), Acidification (AP), Eutrophication (EP) and Global Warming Potential (GWP). The results were then expressed as a quantification of the potential contribution of materials to each impact category.

3. Results and discussion

3.1. Household digesters performance at high altitude

Firstly, small-scale digesters were compared from a technical point of view. The performance of fixed dome and plastic tubular models implemented at high altitude in developing countries is summarised in Table 1. Literature results show how the biogas production rate may range between 0.03 and 0.7 m³ biogas m⁻³ digester d⁻¹, depending on the feedstock (i.e. cow, guinea pig and sheep manure) and operational parameters (i.e. temperature, HRT and OLR). Considering only digesters treating cow manure, among the most common feedstocks in rural zones of developing countries, the biogas production rate may range between 0.07 and 0.47 m³ biogas m⁻³ digester d⁻¹ in plastic tubular digesters adapted to Andean conditions [12,24]; and between 0.35 and 0.7 m³ biogas m⁻³ digester d⁻¹ in fixed dome digesters [8,28–30]. In all cases the methane content in biogas was about 55–60%. The

different location of fixed dome and tubular digesters (1000 vs. 2800–3400 m.a.s.l), feedstock composition and operational parameters (i.e. HRT, OLR) may account for differences on the biogas production rate.

Kanwar and Guleri [8] compared Chinese fixed dome and plastic tubular digesters (without greenhouse) implemented in India at an altitude of 1300 m.a.s.l. In the plastic tubular model, the biogas production rate dropped by 70% during winter months as compared to summer months; while in the fixed dome model the biogas production rate decreased by only 16%. Due to the lack of insulation, in the plastic tubular digester there was a wide temperature fluctuation, which reduced the average temperature of the system. As a result, the biogas production rate was about 30% lower in the plastic tubular model than in the fixed dome model.

Plastic tubular digesters were adapted to Andean conditions by providing digesters insulation and substituting a simple roof used in Tropical regions by a greenhouse. In this manner, the liquor temperature remained fairly constant at around 20 °C [24,31]. Nevertheless, the systems remained within the psychrophilic temperature range, and produced less biogas than mesophilic ones, i.e. digesters located in Tropical regions (>1 m³ biogas m⁻³ digester d⁻¹)

Table 3

Main construction materials for household digesters located at high altitude and cost over a lifespan of 20 years.

Fixed dome – Chinese model		Plastic tubular – Taiwanese model	
Materials	Price (\$)	Materials	Price (\$)
Bricks	806	Bricks	23
Cement	418	Cement	8
Sand and Gravel	21	Sand	5
Metals (iron rods, screws, nails)	110	Metals (iron rods, screws, nails)	174
		Geomembrane – digester (PVC)	1247
		Plastic – greenhouse (LDPE)	117
Plastic and accessories for biogas pipelines and cookstove (HDPE, PVC and clay)	77	Plastic and accessories for biogas pipelines and cookstove (HDPE, PVC and clay)	21
Manpower	532	Manpower	134
Total cost	1963	Total cost	1729

Note: PVC: polyvinyl chloride; LDPE: low-density polyethylene; HDPE: high-density polyethylene.

Table 2

Design and operation parameters of household digesters located at high altitude.

Parameter	Fixed dome – Chinese model	Plastic tubular – Taiwanese model
Feedstock	Cow manure	Cow manure
Influent (% TS)	6–8 [8,24]	6–8 [8,24]
Total and useful volume (m ³)	10 and 8 [23]	10 and 7.5 [24]
Temperature (°C)	<25 [24]	<25 [24]
Hydraulic residence time (days)	60–90 [24]	60–90 [24]
Organic loading rate (kgVS m ⁻³ digester d ⁻¹)	1 [8,24]	1 [8,24]
Biogas production rate (m ³ biogas m ⁻³ digester d ⁻¹)	0.12 [24]	0.12 [24]

Table 4

Inventory analysis of household digesters implemented at high altitude. Values are referred to the functional unit (1 m³ of biogas).

Materials	Fixed dome model (kg FU ⁻¹)	Plastic tubular model (kg FU ⁻¹)
Bricks (sand–lime)	1.16	0.03
Cement	0.35	7.20E–03
Sand	1.24	0.70
Gravel	0.86	–
Metals (iron rods, screws, nails)	0.01	2.30E–03
Plastic and accessories for biogas pipelines (HDPE and PVC)	1.04E–03	7.80E–03
Geomembrane – digester (PVC)	–	2.40E–02
Greenhouse and biogas reservoir (LDPE)	–	7.20E–02
Cookstove (clay)	1.00E–03	1.10E–03

Note: HDPE: high-density polyethylene; PVC: polyvinyl chloride; LDPE: low-density polyethylene.

[10,21]. In this context, the codigestion of manure with other local waste constitutes an alternative to improve the biogas production rate [24].

Regarding the construction, operation and maintenance of household biogas plants, the construction of fixed dome digesters is labour intensive and requires skilled supervision [21]. Construction materials are not always available in rural and remote communities, but they are generally available in larger nearby communities. Plastic tubular digesters are characterised by the ease of implementation and handling. Most importantly, they do not require specialised manpower. On the other hand, high quality prefabricated plastic bags might not be locally available and require importation. Nevertheless, all construction materials can be easily transported.

The operation and ordinary maintenance of both plants does not require skilled operators, but households should be trained to understand the system [32]. Both digesters should be fed every day and digestate should be correctly managed, disposed or reused in agriculture as fertilizer [18]. De-sludging of accumulating solids in the bottom of reactors is required on exceptional basis only. Both plants should also be regularly checked for biogas leakage in the reactor or pipeline. Plastic bags are susceptible to mechanical damage and have a short operational life, typically <5 years [21]. Extraordinary maintenance might be needed if cracks appear in fixed dome digesters, as a result of atmospheric temperature fluctuation or earthquake.

On the whole, plastic tubular digesters should be preferred in remote rural communities, where construction materials transport might be problematic and specialised manpower not easily available.

3.2. Economic analysis

The economic assessment was based on digesters capital cost, 1963 \$ for the fixed dome model and 1729 \$ for the plastic tubular model (Table 3). Bricks and cement accounted for 60% of the total cost in the fixed dome digester. On the other hand, the geomembrane was the most expensive material in the plastic tubular digester (around 70% of the total cost). Manpower was about 30 and 7% of total cost in the fixed dome and plastic tubular digester, respectively.

A lifespan of 20 years was considered for all materials, except for plastic, which was reduced to 5 years in accordance with manufacturers' specifications and Nzila et al. [21]. In this way, the capital cost of the tubular digester included 4 times the geomembrane and greenhouse plastic. The initial investment cost would then be 706 \$ (341 \$ for the geomembrane and greenhouse plastic, plus 365 \$ for

the rest of materials), which represents 36% of the fixed dome digester investment cost. However, the tubular model would require an investment of another 341 \$ every 5 years.

Recently, a new fixed dome model (Camartec) has been developed to minimize construction materials and implementation costs with respect to the traditional Chinese model, by reducing the size of the main chamber and making a second compensation chamber [32]. Up to date, this model has been mainly implemented in Africa [21]. Some authors have estimated that, the capital cost of the Camartec model could be some 50% lower compared to the Chinese model in the Peruvian Andes context [23].

Small-scale digesters in the Peruvian Andes are mainly implemented in rural households whose economy is based on subsistence agriculture (self-sufficient farming), and whose income is lower than 500 dollars per month [19]. Carbon emission trading might be considered in large-scale implementation programmes, aimed at co-financing digesters to make them affordable for rural households.

3.3. Life cycle assessment

Potential environmental impacts of fixed dome and plastic tubular biogas plants located at high altitude are shown in Fig. 2. The individual contribution of each material is represented separately, in order to identify the most relevant issues of each scenario. The distribution of the global impact between different materials (in %) is summarised in Table 5.

The fixed dome model showed the lowest impact in all categories (Fig. 2), appearing as the best alternative from an environmental point of view. The highest impact of the tubular digester was attributed to the plastic employed for the greenhouse and biogas reservoir construction (55–90% depending on the category), followed by the geomembrane (10–35% depending on the category), while the other items accounted for less than 10 % of the total impact in each category (Table 5). In the fixed dome digester, the main impact was attributed to cement (~50% in all categories) and bricks (~30% in all categories) used for the dome construction (Table 5).

In the following categories, Abiotic Depletion, Acidification and Eutrophication, the potential impact of the plastic tubular digester was between 1.5- and 7-fold that of the Chinese model (Fig. 2a–c). This means that the impact of construction materials was much higher for the tubular digester (i.e. plastic and geomembrane) than for the fixed dome digester (i.e. cement and bricks).

However, in terms of Global Warming Potential both systems presented similar results (0.24 and 0.26 kg CO_{2eq} m⁻³ biogas) (Fig. 2d), meaning that total CO₂ emissions of construction materials were equivalent for the fixed dome (i.e. cement and bricks) and tubular (i.e. plastics and geomembrane) digesters. Even though this study was mainly focussed on the construction phase, it is worthwhile mentioning that global warming may decrease as a result of digesters operation. Indeed, biogas combustion produces carbon dioxide, a GHG 21 times less powerful than methane, released when animal manure is left in the fields without any treatment [19,21,33]. Moreover, biogas combustion generates less GHG emissions than traditional biomass, such as firewood, which is generally used by rural households [19,34,35]. Nzila et al. [21] compared the LCA of plastic tubular and Chinese digesters construction and operation in Kenya, observing a GWP reduction of about 1 kg CO_{2eq} after the implementation of both models. Previous studies which only considered digesters operation, reported a CO_{2eq} emission reduction between 4 and 9 tons per year per household digester in temperate and tropical regions, respectively [36,37]. A CO_{2eq} emission reduction of 2.7 tons per year per household was observed for plastic tubular digesters implemented in the Peruvian Andes [19]. Certainly,

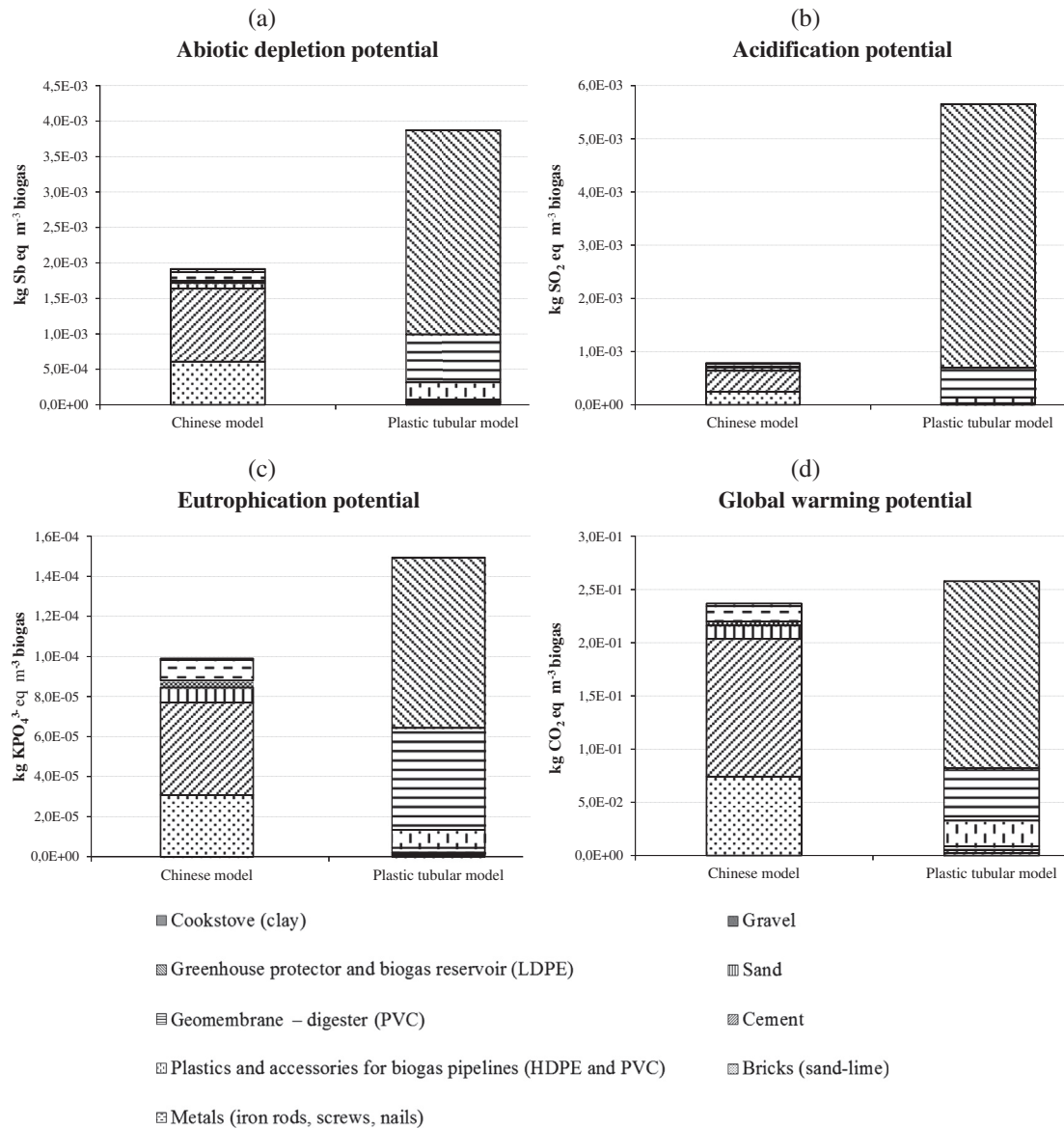


Fig. 2. Comparison of potential environmental impacts of household digesters (fixed dome – Chinese model and plastic tubular – Taiwanese model) per m³ of biogas.

the reduction of CO₂ emissions depends on the biogas production rate, which is more favourable in tropical and temperate regions.

Household tubular digesters are generally regarded as an environmentally sustainable technology [9,19,21]. Indeed, they reduce GHG emissions and deforestation by replacing firewood and traditional biomass [19,38]. Comparing the studied models, fixed dome digesters appeared to be more environmentally friendly as a result of construction materials. The highest impact of the tubular digester may be explained by the type (plastics vs. bricks and cement) and lifespan (5 years for plastics vs. 20 years for concrete and bricks) of construction materials. In order to improve the environmental performance of tubular digesters, the use of plastics should be minimised, the lifespan of construction materials expanded and more environmentally respectful materials used. For instance, bioplastics appear as potential solution to reduce the environmental impact of this system. Bioplastics present several advantages over conventional petroleum-based plastics, such as decreased CO₂ emissions and lower energy consumption [39,40]. Nevertheless, the economic and environmental sustainability of

bio-based plastics should be studied in depth before replacing current materials [41].

4. Conclusions

This study compared two household digester models implemented at high altitude. Over a lifespan of 20 years, the capital cost was 12% lower for the plastic tubular digester compared to the fixed dome digester. Moreover, the initial investment cost of the tubular digester was 1/3 that of the fixed dome digester, although it would require some extra investment every 5 years to replace plastic materials. The LCA showed how the plastic tubular digester caused the highest impact; as a result of the relatively short lifespan of plastic materials and geomembrane. In the fixed dome model, most environmental impact corresponded to concrete and bricks. The main advantage of the tubular digester was its ease of implementation and handling, and lower investment cost compared to the fixed dome digester, which appeared to be more environmentally friendly.

Table 5

Contribution of construction materials to the potential environmental impact (in %) of household digesters at high altitude.

Construction materials	Abiotic Depletion		Acidification		Eutrophication		Global Warming Potential	
	Fixed dome model	Plastic tubular model	Fixed dome model	Plastic tubular model	Fixed dome model	Plastic tubular model	Fixed dome model	Plastic tubular model
Bricks (sand–lime)	31.76	0.47	31.35	0.13	31.16	0.62	31.22	0.86
Cement	53.98	0.56	50.92	0.15	46.71	0.64	54.74	1.05
Sand	4.22	0.05	7.33	0.03	7.53	0.13	5.25	0.13
Gravel	1.33	0.00	2.55	0.00	3.53	0.00	1.57	0.00
Metals (iron rods, screws, nails)	6.52	0.76	6.50	0.21	10.42	1.62	6.12	1.32
Plastic and accessories for biogas pipelines (HDPE and PVC)	2.19	6.33	1.33	1.98	0.65	5.89	1.09	9.32
Geomembrane – digester (PVC)	0.00	17.48	0.00	9.89	0.00	34.11	0.00	19.10
Greenhouse and biogas reservoir (LDPE)	0.00	74.35	0.00	87.62	0.00	56.98	0.00	68.22
Cookstove (clay)	0.01	0.00	0.01	0.00	0.01	0.01	0.01	0.01
Total impact	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Note: HDPE: high-density polyethylene; PVC: polyvinyl chloride; LDPE: low-density polyethylene.

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